Explicit Disk-Algebra Cauchy Transforms under Log–Log Entropy

A substantial partial answer to Limani–Malman

Automated Mathematics Run `fa_banach_001`, lane 4

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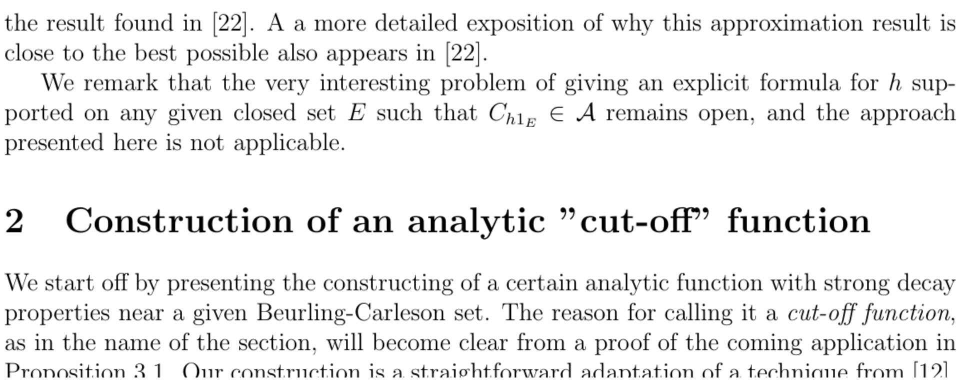
Status. `candidate_substantial_partial_likely_valid`. An explicit formula is obtained for a class strictly larger than the Beurling–Carleson class. The problem for an arbitrary closed set remains open.

1 Source problem

In the Introduction of arXiv:2111.14112v3, PDF page 6, Limani and Malman ask for an explicit formula for a nonzero measurable function h supported on a given closed set $E \subset \mathbb{T}$ such that

$$C_h(z) = \int_E \frac{h(\zeta)}{1 - z\overline{\zeta}}\, dm(\zeta)$$

belongs to the disk algebra $\mathcal{A}$ [1]. Their construction gives the stronger conclusion $C_h \in \mathcal{A}^\infty$ when E is a positive-measure Beurling–Carleson set.


the result found in [22]. A a more detailed exposition of why this approximation result is close to the best possible also appears in [22].

We remark that the very interesting problem of giving an explicit formula for h supported on any given closed set E such that $C_{h1_E} \in \mathcal{A}$ remains open, and the approach presented here is not applicable.

2 Construction of an analytic ”cut-off” function

We start off by presenting the constructing of a certain analytic function with strong decay properties near a given Beurling-Carleson set. The reason for calling it a *cut-off function*, as in the name of the section, will become clear from a proof of the coming application in Proposition 3.1. Our construction is a straightforward adaptation of a technique from [12]

Figure 1: The open statement in the Introduction of the source paper, PDF page 6.

We assume throughout that E is proper and has positive normalized Lebesgue measure; these are necessary for a nonzero integrable density supported on E. The case $E = \mathbb{T}$ is trivial.

2 Statement and proof intuition

Let $\{I_n\}$ be the component arcs of $\mathbb{T} \setminus E$, and set

$$\Phi(t) = \log\bigl(e + \log(1/t)\bigr), \qquad 0 < t \le 1.$$

The source construction places poles just outside Whitney subarcs and weights a subarc of length ℓ by $\ell\log(1/\ell)$. This pays enough entropy to make the cutoff flat to every order. For the disk algebra, flatness is not needed. We instead use the smaller weight $\ell\Phi(\ell)$. The cutoff then decays like a chosen negative power of $\log(e/\operatorname{dist}(\cdot,E))$.

On the complement of E, conjugate the cutoff and extend it by zero across E. A two-scale estimate shows that the extension has modulus of continuity

$$\omega(\delta) \lesssim \left(\log(e/\delta)\right)^{-p} + \delta^{1/3}, \qquad p > 1.$$

This is a Dini modulus. The classical Dini theorem makes its harmonic conjugate continuous, hence its analytic projection lies in $\mathcal{A}$. Finally, the untruncated conjugate cutoff is anti-analytic, so its analytic projection vanishes. Splitting the circle into E and its complement gives the desired Cauchy transform supported on E.

Theorem 1 (Log–log entropy construction). *Let $E \subsetneq \mathbb{T}$ be closed with $m(E) > 0$. Suppose its complementary arcs satisfy*

$$\sum_n |I_n|\,\Phi(|I_n|) < \infty. \tag{1}$$

Then there is an explicit bounded nonzero function h, supported on E, such that $C_h \in \mathcal{A}$.

3 The explicit formula

Perform the Whitney decomposition from Lemma 2.1 of [1]:

$$I_n = \bigcup_{k\in\mathbb{Z}} B_{n,k}, \qquad \ell_{n,k} := |B_{n,k}| = \frac{|I_n|}{3\cdot 2^{|k|}},$$

where $\ell_{n,k} = \operatorname{dist}(B_{n,k}, E)$. Relabel these arcs as B_j, their lengths as ℓ_j, and their midpoints as $b_j \in \mathbb{T}$. Put $r_j = 1 + \ell_j$.

For a fixed numerical $K > 0$ chosen sufficiently large, define

$$F_E(z) = -K\sum_j \frac{b_j\ell_j\Phi(\ell_j)}{r_jb_j - z}, \qquad g_E(z) = \exp(F_E(z)), \qquad z \in \mathbb{D}. \tag{2}$$

If g_E^* denotes the almost-everywhere radial boundary value, set

$$h_E(\zeta) = 1_E(\zeta)\,\overline{\zeta g_E^*(\zeta)}. \tag{3}$$

Equations (2)–(3) are the promised formula.

4 Proof

Lemma 2 (Whitney summability). *Condition* (1) *implies*

$$\sum_j \ell_j\Phi(\ell_j) < \infty.$$

Proof. For $L = |I_n|$, the contribution of its Whitney pieces is

$$\frac{L}{3}\Phi(L/3) + 2\sum_{k\geq 1}\frac{L}{3\cdot 2^k}\Phi\left(\frac{L}{3\cdot 2^k}\right).$$

Since $\Phi(L/(3\cdot 2^k)) \leq C(\Phi(L) + \log(k+2))$ and both $\sum 2^{-k}$ and $\sum 2^{-k}\log(k+2)$ converge, this is at most $CL(1+\Phi(L))$. Here $\Phi \geq 1$, so summing in n proves the claim. □

By Lemma 2, the series in (2) converges normally on compact subsets of $\mathbb{D}$. Moreover

$$\Re\frac{b_j}{r_j b_j - z} = \Re\frac{1}{r_j - \bar{b}_j z} > 0,$$

so $\Re F_E < 0$ and $|g_E| \le 1$. The function is nonzero in $\mathbb{D}$, and the boundary uniqueness theorem for H^∞ implies $g_E^* \neq 0$ almost everywhere on $\mathbb{T}$. Consequently h_E in (3) is bounded and nonzero.

The poles in (2) cluster only at E, so g_E continues analytically across each compact subarc of $\mathbb{T} \setminus E$. If $\zeta \in B_j$, the jth summand and Whitney geometry give

$$\Re\frac{b_j}{r_j b_j - \zeta} \ge \frac{c}{\ell_j}.$$

All summands have positive real part before multiplication by $-K$; hence

$$|g_E(\zeta)| \le \exp\{-cK\Phi(\ell_j)\} = \big(e + \log(1/\ell_j)\big)^{-cK}. \tag{4}$$

For $\zeta \in B_j$, $\ell_j \asymp \operatorname{dist}(\zeta, E)$. Choosing K so that $p := cK > 2$, equation (4) becomes

$$|g_E(\zeta)| \le C\big(\log(e/\operatorname{dist}(\zeta, E))\big)^{-p}. \tag{5}$$

Define on the circle

$$S(\zeta) = 1_{\mathbb{T}\setminus E}(\zeta)\overline{\zeta g_E^*(\zeta)},$$

using the analytic continuation on $\mathbb{T} \setminus E$ and the value zero on E. We next verify the Dini modulus claimed above. Write $d(\zeta) = \operatorname{dist}(\zeta, E)$. Whitney geometry also gives, uniformly in j,

$$|r_j b_j - \zeta| \ge c\, d(\zeta), \qquad \zeta \in \mathbb{T} \setminus E.$$

Indeed, a Whitney center close to ζ has length comparable with $d(\zeta)$, while every other center is separated from ζ by a constant multiple of $d(\zeta)$. With $M = \sum_j \ell_j \Phi(\ell_j) < \infty$, differentiation of (2) therefore yields

$$|F_E'(\zeta)| \le CKMd(\zeta)^{-2}, \qquad |S'(\zeta)| \le C_E d(\zeta)^{-2}. \tag{6}$$

Let two circle points be at arc distance at most δ. If either is within $2\delta^{1/3}$ of E, then both are within $3\delta^{1/3}$ of E, and (5) bounds their two values by $C(\log(e/\delta))^{-p}$. Otherwise the shorter connecting arc stays at distance at least $\delta^{1/3}$ from E, and (6) plus the mean-value theorem bounds the difference by $C_E\delta^{1/3}$. Thus

$$\omega_S(\delta) \le C_E\left[(\log(e/\delta))^{-p} + \delta^{1/3}\right]. \tag{7}$$

Since $p > 1$, the right-hand side divided by δ is integrable near zero. Hence S is Dini continuous.

The Dini theorem for conjugate functions states that the harmonic conjugate of a Dini-continuous circle function is continuous [2, Chapter III]. Equivalently, its nonnegative-frequency Riesz projection P_+S is the boundary function of an element of $\mathcal{A}$.

Finally, $s := \overline{\zeta g_E^*(\zeta)}$ belongs to $\overline{zH^2}$, so $P_+ s = 0$. Since $s = h_E + S$,

$$C_{h_E} = P_+ h_E = -P_+ S \in \mathcal{A}.$$

If this function were zero, then $h_E \in \overline{zH^2}$; but h_E vanishes on the nonempty open set $\mathbb{T} \setminus E$, so boundary uniqueness would force $h_E = 0$, a contradiction. This completes the proof of Theorem 1. $\square$

5 Strict extension beyond Beurling–Carleson sets

Example 3. *For each sufficiently large integer k, take $N_k = \lfloor ae^k/k^2 \rfloor$ disjoint arcs of length e^{-k}, with $a > 0$ small enough that their total length is less than one, and let E be the complement of their union. Then $m(E) > 0$ and*

$$\sum_k N_k e^{-k}\Phi(e^{-k}) \asymp \sum_k \frac{\log(k+e)}{k^2} < \infty,$$

whereas

$$\sum_k N_k e^{-k}\log(e^k) \asymp \sum_k \frac{1}{k} = \infty.$$

Thus E satisfies Theorem 1 but is not a Beurling–Carleson set. Hence the explicit formula (2) *applies directly to closed sets outside the hypothesis of the source paper's cutoff construction.*

6 Verification, novelty, and limitations

The proof uses only positive-term estimates. The accompanying script checks the convergence/divergence separation in Example 3 and the bounded ratio between the Whitney and original log–log entropy sums. It is a consistency check, not part of the proof.

A bounded literature search through 27 August 2026 used arXiv:2111.14112 and the phrases "explicit Cauchy transform closed set", "Dini Cauchy transform Beurling–Carleson", "log log Beurling–Carleson", and "loglog entropy disk algebra". It found the source article and standard Dini-regular Cauchy-transform results, but no matching weakening of the source's entropy hypothesis. Novelty confidence is moderate, not definitive.

The theorem does not solve the stated problem for every positive-measure closed set. The remaining obstruction is exactly the possible divergence of (1); the present positive exterior-pole construction then cannot simultaneously have finite pole mass and a uniform Dini modulus.

Human-review recommendation. Review as a likely-valid substantial partial result. The highest-value checks are the denominator estimate preceding (6), the two-case Dini modulus proof, and the boundary uniqueness step proving nontriviality.

References

[1] A. Limani and B. Malman, *Constructions of some families of smooth Cauchy transforms*, Canadian Journal of Mathematics **76** (2024), 319–344; arXiv:2111.14112.

[2] A. Zygmund, *Trigonometric Series*, Vol. I, Cambridge University Press, third edition, 2002.